

A Realistic Implementation of Fast Convergence Hypernetwork-based DSP for Coherent Data Centre Links

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Abstract *We experimentally demonstrate a hypernetwork-based algorithm to generate the weights of an adaptive equalizer in 30-GBaud DP-16-QAM coherent transmission. It achieves better performance than conventional algorithms, providing purely feed-forward coherent DSP and locking to unseen channels with comparable computational complexity but without feedback paths. ©2026 The Author(s)*

Introduction

Capacity demands for data centre interconnects (DCI) are increasing exponentially. While IM/DD solutions offer low costs, fast channel acquisition, and low processing latency, they are struggling to meet future demands^[1]. Coherent systems offer much higher spectral efficiency but with increased cost and latency, with the latter being a key technical obstacle in DCI with AI/HPC workloads^[2].

A major source of channel acquisition latency in coherent systems is the use of iterative optimisation in DSP blocks, as these operate at ASIC clock rates much lower than the symbol rate^[3]. Adaptive equalisation compensates stochastic effects such as fibre polarisation-mode dispersion (PMD), typically using iterative algorithms like CMA and DD-LMS^[4]. However, DCI traffic often exhibits burst behaviour^{[5][6]}, and with increasing interest in (fast) optical circuit switching (OCS)^[7], DSP must frequently re-converge. Iterative algorithms may require 1000s of symbols to converge, greatly impacting network performance. Also, they often rely on prior chromatic dispersion (CD) compensation (CDC), adding processing and parameter acquisition overhead^[8], or limited CD tolerance^[9].

Data-aided equaliser acquisition can reduce latency with predictable convergence time^[10]. This still typically relies on iterative refinement, as existing feed-forward options (e.g., least-squares) require a high numerical precision impractical for hardware implementation. This motivates developing alternative low-latency convergence schemes.

We propose a hypernetwork-based approach for adaptive equaliser convergence. Hypernetworks are neural networks that generate the parameters of a target network from a given input context^[11]. Here, a pilot sequence is used as the context to directly generate the coefficients of a MIMO equaliser. This enables feed-forward convergence without iterative optimisation, achieving effective equalisation even with uncompensated CD. Our approach supports low-latency DSP operation on previously unknown channels, enabling flexible networking and fast OCS in coherent systems.

Fast-convergence DSP

To achieve fast-convergence DSP, we use a preamble including a CD-resilient 64-symbol frame marker and a 64-symbol training sequence. The frame marker consists of zeros, followed by three **1** and three **-1** across both polarisations, and trailing zeros. This structure limits CD-induced inter-symbol interference. Synchronisation is performed via a cross-correlation-triggered threshold. Following the frame marker, a 64-symbol CAZAC sequence is used as a pilot for equaliser convergence. Its short length enables rapid channel acquisition with low computational overhead. The same pilot is both for conventional DSP and hypernetwork-based DSP.

As a baseline we use a conventional coherent DSP chain with IQ-deskew, orthogonalisation, frame synchronisation, CDC, matched filter, carrier frequency offset (CFO) estimation and compensation using the full transmitted signal as a reference, adaptive equaliser by carrying out the recursive least squares (RLS) algorithm given the CAZAC sequence, and carrier phase recovery (CPE) using QPSK pilots inserted every 32 symbols. CDC and CFO are typically estimated prior to equalisation, since iterative algorithms may fail to converge with these impairments uncompensated in the adaptation loop. While this can be taken into account^[9], this approach introduces significant locking latency.

Instead of CDC and RLS, we use hypernetwork to generate a MIMO filter, and utilising the same pilot-aided CPE. In addition, as the hypernetwork is trained to output coefficients even in the presence of CFO (as described later), this technique allows for frequency offset estimation after adaptive equalisation. This improves estimates and avoids having to perform coarse CFO compensation before MIMO and residual CFO compensation after MIMO. By reducing inter-stage dependencies and eliminating iterative convergence, this DSP chain enables low-latency channel acquisition suitable for burst-mode and dynamically reconfigured optical networks.

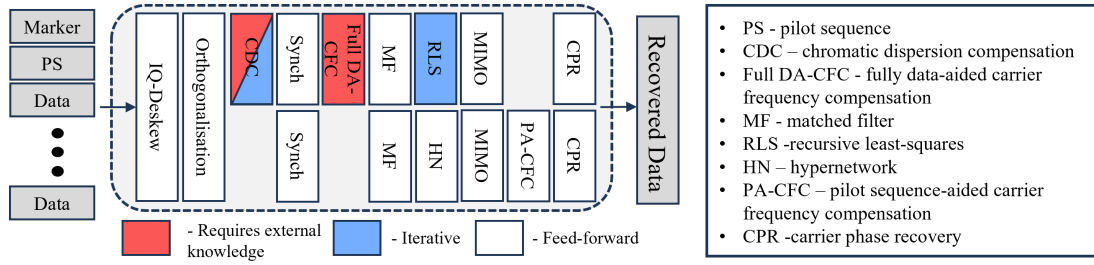


Fig. 1: Conventional (top branch) and hypernetwork-based (bottom branch) DSP.

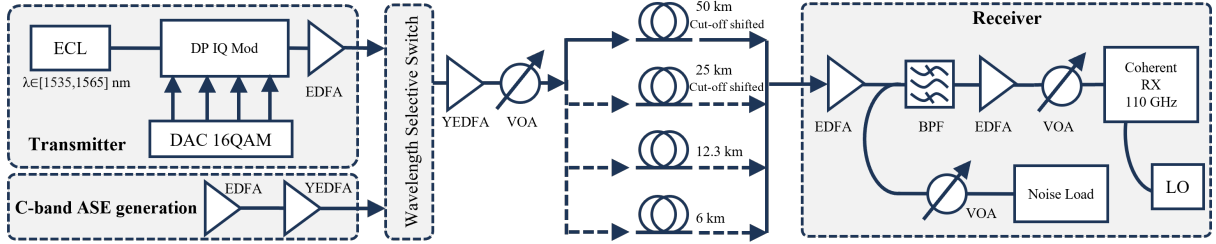


Fig. 2: Experimental transmission setup. A 100 kHz linewidth tunable external cavity laser (ECL) is modulated using a dual-polarisation IQ modulator (DP IQ Mod) with a 90 GSa/s DAC. A wavelength selective switch (WSS) is used to shape ASE into a C-band flat spectrum, carve a notch and multiplex in the modulated channel - since YEDFAs are used the first usable wavelength is 1540nm. A variable optical attenuator (VOA) controls launch power, where the signal is launched into a variety of fibres via a switch. The output is amplified, coupled with noise loading, and passed through a bandpass filter (BPF) implemented by another WSS to acquire the modulated. This is then received at a 110 GHz coherent receiver, with a 100 kHz linewidth local oscillator (LO).

Fig. 1 outlines the DSP chain for both hypernetwork-based and the conventional approach used in this paper. While the conventional approach relies on external knowledge and iterative procedures (CDC estimation can be carried out with either approach), the hypernetwork-based DSP allows for a fully feed-forward implementation, even when locking to unknown channels.

Hypernetwork Training

To acquire a hypernetwork that is capable of producing valid adaptive filters over a range of experimental channels, it must first undergo an extensive training phase over a range of simulated channels similarly to the training method in^[12]. This is performed with a linear transmission simulation that takes into account CD, PMD, CFO, laser phase noise, clock delay, and AWGN. A 64-symbol CAZAC sequence is then transmitted under these conditions to act as the known pilot sequence, which is then inputted to a hypernetwork to output the MIMO filter. A data payload is then sampled from a Gaussian constellation and is transmitted using the same Jones-space transfer matrix as for the pilot sequence, with other impairments ignored. The MIMO filter then attempts to recover the transmitted symbols, and the mean-squared error is computed and used to propagate gradients to train the hypernetwork. In this way, the hypernetwork should learn to output a MIMO equaliser that is the inverse of the channel. This is simulated over a range of parameters such that the hypernetwork receives coverage over the expected range of channel conditions. This hypernetwork is trained until the loss converges.

The hypernetwork itself is implemented as a simple multi-layer perceptron (MLP). The incoming pilot sequence is first decomposed into its real X , Y , I , and Q components, and then flattened to be given to the MLP as an input vector of size 512 (128 samples with four tributaries). The MLP has 4 hidden layers of size 1024, using the ReLU activation. The output is then a vector of size 88, representing the real and imaginary coefficients of the 11-tap 2×2 MIMO filter.

Once trained, the hypernetwork undergoes model complexity reduction techniques to improve computational complexity. This includes structured pruning^[13] of 50% of neurons and clustering the weights^[14] of each neuron to 128 centroids. This reduces computational complexity by 16-times (in terms of real multiplications) without loss of performance.

Experimental Setup

The transmission set up is shown in Fig. 2. We carry out transmission over various lengths of single-mode fibre, 6 km (G.652.B), 12.3 km (G.652.B), 25 km (G.654.B/D), and 50 km (G.654.B/D), and every concatenated combination thereof. A ~ 100 kHz linewidth laser is used for transmission. Transmission signal consists of frame marker, CAZAC pilot sequence, and a DP-16-QAM data payload of 2^{16} symbols with QPSK pilots inserted every 32 symbols. RRC pulse shaping with a rolloff of 0.1 is applied. Intradyne detection is used given an additional ~ 100 kHz linewidth local oscillator, and additional noise loading to limit the impact of transceiver drift on the results. Furthermore, C-band shaped ASE is added at the

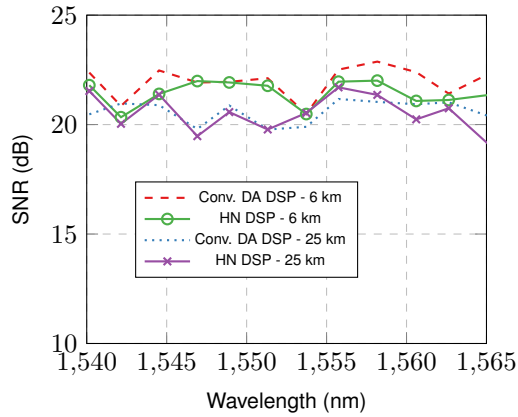


Fig. 3: Maximum SNR against transmission wavelength for varying fibre lengths, for Hypernetwork (HN), conventional (Conv.), and conventional data-aided (Conv. DA) DSP. Data points averaged over 50 frames.

transmitter to emulate a fully populated spectrum.

Results

To investigate suitability for the hypernetwork-based DSP over the C-band Fig. 3 shows the performance (constellation SNR) of both conventional DSP and hypernetwork-based DSP over various transmission wavelengths; only 6 km and 25 km shown for readability. It can be seen that both techniques are able to achieve similar performance, even with the lack of iterative convergence for the hypernetwork.

To further analyse performance, Fig. 4 shows SNR against launch power at a transmission wavelength of 1553.7 nm for 6 km and 25 km. Again, the hypernetwork is able to achieve similar performance to the conventional data-aided DSP over multiple distances – similar observations for all distances up to 60 km.

Lastly, Fig. 5 shows the performance of both techniques at optimal launch power over fibre distance. The hypernetwork is able to outperform the conventional approach for length < 40 km, whilst the relative performance drops significantly from 60 km. This is due to the fact that the MIMO filter outputted by the hypernetwork is limited to 11 taps, and the CD after 50 km begins to require additional FIR length for full compensation. Whilst the lack of CDC remains an advantage for hypernetworks due to simpler channel locking, other approaches would need to be considered (such a O-band or hollow core fibre transmission) for longer reaches.

Discussion and Conclusion

In this paper we introduced a hypernetwork-based framework for adaptive equaliser convergence over DCI distances. This framework enabled feed-forward convergence of the equaliser in the presence of uncompensated CD and CFO. This allows for lower latency by bypassing CDC, and leaving CFO estimation and compensation until after equalisation. This can be utilised for extremely fast

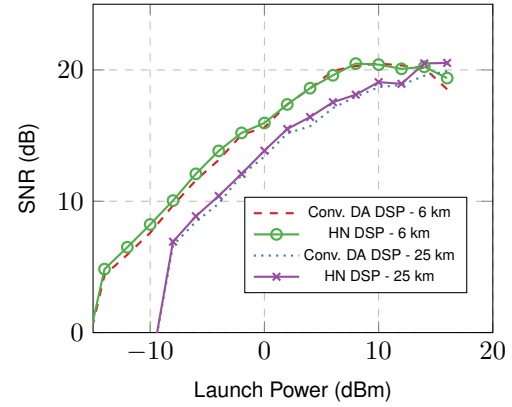


Fig. 4: Experimental SNR against launch power for varying fibre lengths at 1553.7 nm, for Hypernetwork (HN) and conventional data-aided (Conv. DA) DSP. Data points averaged over 50 frames.

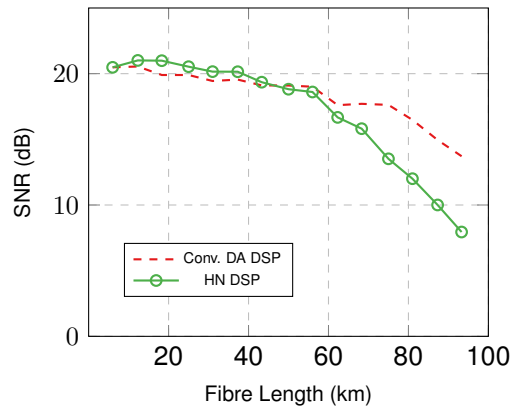


Fig. 5: SNR against fibre length at the optimal launch power at 1553.7 nm. Data points averaged over 50 frames.

channel locking within DCIs, enabling techniques such as ultra-fast OCS, without compromising on achieved SNR.

Meanwhile, the computational complexity footprint of the hypernetwork remains low. As implemented here, with pruning and weight clustering, it can be evaluated with $\sim 300k$ real multiplications. As these multiplications are static (the hypernetwork weights are static), they can be implemented as multiplierless constant multiplications^[15], allowing for extremely efficient hardware implementation. This is compared with standard iterative approaches, which have similar computational complexity in terms of real multiplications, but require more expensive MAC units to compute on hardware given that it multiplies two arbitrary quantities.

It should also be noted that RLS was used as the conventional algorithm of choice to converge the equaliser. This is due to the fact that LMS struggles to converge with short training sequences, and therefore would require additional locking latency. Meanwhile, while RLS provides reliable convergence with short training sequences, it is impractical to implement it in hardware given its sensitivity to numerical precision^[16]. Thus, the hypernetwork-based approach supersedes both algorithms in both hardware feasibility and latency.

Acknowledgements

This work was supported by a UKRI Future Leaders Fellowship under grant no. MR/Y034260/1, by a Strategic Equipment under Grant no. EP/V007734/1, by a EPSRC Programme grant under Grant no. EP/R035342/1 and by an EPSRC Doctoral Studentship under Grant no. EP/T517793/1 and EP/W524335/1. Underlying data is accessible at doi.org/10.5522/04/32080557.

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